

Temporally Constrained, Provenance-Verified Evidence Retrieval for Indian Legal Research

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Abstract—AI-assisted legal research fails in ways that fluent output conceals: a cited authority may not exist, may not contain the attributed passage, may post-date the matter under analysis, or may never have been retrieved at all. A 2026 Supreme Court of India ruling overturned tribunal findings that had relied on exactly these defects, turning citation verifiability from a matter of style into a functional requirement for any such system. This paper presents an evidence pipeline for historical Indian Supreme Court judgments in which every displayed citation is temporally eligible, provenance-bound, and mechanically verifiable against the persisted corpus. The design is deliberately extractive: material propositions are shown verbatim with page and character locators, and outcome prediction is kept in a separate field so that a classification score can never stand in for evidence quality. We report three findings on a combined 37-case reference-evidence population (a frozen 30-case base plus a later additive 7-case verified extension, analysed in strata throughout). First, a cross-corpus audit shows that naive identifier matching between the ILDC corpus and an eCourts-derived judgment archive is unsafe: of 5,391 syntactically matching identifiers only 11 survived content alignment, motivating a reusable content-alignment gate. Second, moving temporal eligibility into the BM25 candidate relation before ranking, together with salient-term query construction and a coverage-qualified self-match guard, raised expected-authority Recall@5 from 5/30 to 12/30 and Recall@100 from 12/30 to 15/30 on the frozen base; on the combined 37 cases Recall@5 is 12/37 and Recall@100 is 20/37. Third, and centrally, all 185 displayed citations passed grounding, provenance, and temporal verification, with zero unsupported claims across the 37 cases, while 17 of the 37 expected authorities remained absent at $k=100$. Perfect verification of what a system displays is therefore compatible with substantially incomplete retrieval, and the two must be measured on separate denominators. Auxiliary outcome baselines on 1,503 held-out ILDC cases and an inference-time evidence-augmentation extension are reported with their own populations and stated limits. A blinded 14-case, four-LLM-rater exploratory comparison favoured the structured presentation 56/56, which we report as presentation evidence only, not human-subject evidence.

Index Terms—legal information retrieval, explainable AI, provenance verification, temporal filtering, retrieval-augmented systems, Indian legal NLP, BM25, InLegalBERT

I. Introduction

Legal research operates in a domain where fluency is not evidence. A generated response can be linguistically convincing while citing a judgment that does not exist, attributing a passage to a real judgment that does not

contain it, or relying on law that was unavailable at the time the matter arose. Empirical profiling of legal hallucination in large language models has established that these failures are frequent rather than exceptional [13], and related work has probed how far explicit legal standards can be communicated to language models at all [14].

The problem is no longer confined to the literature. In *Pooja Ramesh Singh v. Jammu and Kashmir Bank Ltd. & Anr.*, 2026 INSC 668, the Supreme Court of India set aside tribunal decisions after identifying nonexistent authorities, incorrect citations, and fabricated passages apparently produced through AI-assisted research [6]. The Court distinguished legitimate assistance from unverified reliance and emphasised continued human control over adjudication. The judgment converts an accuracy concern into a systems requirement: a legal research tool must produce evidence that a reviewer can locate, inspect, and independently verify, not merely an answer that reads as legal.

This paper treats that requirement as the primary design constraint. We build and evaluate an evidence pipeline over historical Indian Supreme Court judgments in which the answer layer is extractive by construction. Given facts extracted from a query judgment, the system retrieves earlier Supreme Court material, retains stable source and passage locators through every stage, excludes the query judgment and its content-aligned duplicates, renders only selected verbatim passages, and then verifies each displayed citation against the persisted corpus and the recorded retrieval run. Outcome prediction is available but structurally subordinated: it is emitted in a separate top-level field and never alters the evidence-bound conclusion.

The central empirical observation is a dissociation that a single aggregate score would erase. Under the final configuration, all 185 displayed citations across the combined 37 cases cleared grounding, provenance, duplicate, and temporal checks, with no unsupported claim among the 37 cases; at the same time, 17 of the 37 independently verified expected authorities never appeared among the top 100 candidates returned by the retriever. Correctness of what is shown and completeness of what is found are therefore separate properties, and a system can score perfectly

on the first while still missing outcome-relevant material on the second. Reporting only a combined “citation correctness” figure would hide exactly the gap a reader needs in order to decide how much to trust the output.

Contributions. This work makes four contributions, each bounded to the evaluated setting.

- 1) A quantified cross-corpus identity failure and its remedy. Converting eCourts identifiers of the form YYYY INSC N to ILDC-style YYYY_N produced 5,391 syntactic candidates, of which only 11 passed content alignment (Section IV-C). We show that identifier equality is unsafe for this widely used dataset pair and supply a reusable content-alignment gate based on title/party identity plus direct phrase overlap.
- 2) Pre-ranking temporal eligibility. We implement historical availability as a predicate inside the BM25 candidate relation rather than as a post-hoc filter, so that ineligible judgments cannot consume retrieval depth (Section VI-D). On the held-out evidence set this change moved Recall@5 from 5/30 to 12/30 without regressing any prior success.
- 3) A four-state citation protocol with separate denominators. We distinguish fabricated citations, real-but-unretrieved citations, retrieved citations violating policy, and valid citations that simply differ from the reference authority (Section VII) — four states that a single correctness label collapses.
- 4) An evaluation that keeps recovery and integrity apart. We report authority recovery, citation integrity, outcome prediction, and explanation presentation on their own populations, and document the resulting dissociation rather than resolving it into a headline number (Section VIII).

Scope. The claims are deliberately narrow. We do not establish legal correctness for retrieved authorities, deploy an autonomous legal advisor, or show that evidence retrieval improves outcome prediction. The system is a researcher-facing prototype evaluated on four distinct and non-pooled populations: 1,503 outcome cases, 30 frozen plus 7 additive source-verified evidence cases analysed in strata, and paired explanation examples (seven formative author-reviewed pairs, superseded by a 14-case exploratory LLM comparison).

II. Related Work

A. Indian legal NLP and judgment prediction

The ILDC corpus established Indian Supreme Court judgment prediction and explanation as a benchmark task [1], and supplies the fixed splits and binary outcome labels used for the baselines here. Domain-adapted pre-training for Indian law produced InLegalBERT and related models [2], which motivate our neural baseline. Work on long Indian judgment summarisation with practitioner evaluation [12] established that document length and

expert-facing presentation are first-order concerns in this jurisdiction. None of these lines addresses cross-corpus source identity or per-citation provenance, which is where this work sits.

B. Legal retrieval, entailment, and benchmarks

COLIEE evaluates case-law and statute retrieval and entailment [9]; LegalBench measures multiple forms of legal reasoning in large language models [10]; CaseHOLD tests selection of cited-case holdings and the value of legal-domain pre-training [11]. These benchmarks set a useful bar for retrieval quality and evaluation rigor, but each operates over one fixed, curated corpus and none requires a candidate to have existed, or been decided, before the query matter arose.

C. Temporal and provenance constraints

TaxFlow applies hybrid RAG with temporal filtering to Indian statutory tax question answering [3], sharing our concern that legal retrieval must respect changing law. CaseFacts frames U.S. legal claim verification with Supported/Refuted/Overruled labels [4], demonstrating that legal validity evolves. LexTime benchmarks temporal ordering of events in U.S. federal complaints [5]. Retrieval-augmented generation established retrieval as an explicit non-parametric evidence source [15]; BM25 supplies the sparse ranking basis [7] and Tesseract the OCR engine for corpus repair [8].

D. Positioning

Table I states the boundary of the present contribution against the closest work. We claim neither the first legal RAG system, nor the first Indian legal model, nor the first temporal legal benchmark. The contribution is the reproducible combination of cross-corpus content-alignment gating, conservative pre-ranking eligibility, exact passage and retrieval-run verification, and separate reporting of authority recovery versus citation validity against a source-first answer key. Because jurisdictions, tasks, corpora, and denominators differ across these systems, none is used as a directly comparable numerical baseline; we compare only experiments run under our own frozen populations.

III. Problem Formulation

Let a historical query judgment have identifier q , query year Y_q , and frozen facts-only text F_q . Let each candidate evidence passage p belong to a source judgment with a stable source identifier, an exact decision date, citation metadata, page and character boundaries, and stored passage text.

The temporal policy admits p only when the decision year of its source is strictly earlier than Y_q . A same-year source is treated as ambiguous, because ILDC supplies no exact query date; a source with missing or unparseable date metadata is excluded. Both rules are deliberately

TABLE I

Positioning against the closest related work. No external system is used as a numerical baseline, because jurisdictions, tasks, corpora, and denominators differ.

Work	Task and setting	Relation to this work	Boundary of present contribution
ILDC / CJPE [1]	Indian Supreme Court outcome prediction and explanation	Supplies the prediction setting and the fixed ILDC Single splits	We use a fixed ILDC subset for baselines and add a separate, provenance-bearing evidence corpus; we do not reproduce the CJPE architecture
InLegalBERT [2]	Domain-adapted pre-training for Indian legal tasks	Motivates the pinned neural outcome baseline	We correct long-document truncation via chunk-and-pool and do not assume domain pre-training must beat a sparse baseline
TaxFlow [3]	Hybrid RAG with temporal filtering for Indian statutory tax QA	Shares the premise that legal retrieval must respect changing law	We address historical precedent retrieval, cross-corpus identity, exact displayed-passage provenance, and a source-verified authority key
CaseFacts [4]	U.S. legal claim verification and precedent retrieval	Shows legal validity evolves and unrestricted retrieval admits noisy sources	We query from historical Indian case facts and verify source identity, temporal eligibility, and retrieval provenance rather than claim labels
LexTime [5]	Temporal ordering of events in U.S. federal complaints	Establishes temporal reasoning as a distinct legal-language challenge	Our task is not event ordering; we enforce historical availability through metadata before ranking

conservative: they trade recall for the guarantee that no displayed authority can post-date the matter.

The evidence task decomposes into three separable questions.

- Recovery. Does the top-100 candidate set contain the independently verified expected authority, and is that authority among the five displayed sources?
- Integrity. Does each displayed citation reproduce a real retrieved corpus passage with exact provenance, avoid the query judgment and its aligned duplicates, and satisfy the temporal rule?
- Presentation. Given identical evidence and citations, does a structured Issue–Authority–Evidence–Conclusion–Uncertainty format make the material easier to inspect than unstructured prose?

Outcome prediction is a secondary evaluation capability, not a fourth research question. It is reported on its own population and never pooled with the evidence measures, so that a classification score cannot substitute for evidence quality and a traceable citation cannot be mistaken for complete authority recovery. Table II fixes the operational definitions and pass/fail criteria.

A. Research questions

There are exactly three research questions. RQ1: Does grounding an Indian legal AI workflow in retrieved legal evidence improve legal-research reliability and relevant-evidence retrieval compared with a facts-only legal-language baseline? The controlled comparison is E2 versus E3; E1 is a traditional baseline reported for context. RQ2: Does provenance-constrained evidence selection and

citation verification reduce unsupported or unverifiable legal claims in the final output? RQ3: Can the proposed structured explanation format improve human-verifiable transparency without materially degrading prediction or retrieval performance?

IV. Corpus Construction and Auditing

A. Purpose-split corpus design

We use two corpora for different experimental purposes rather than treating them as interchangeable records of the same judgments.

ILDC Single supplies fixed case-level splits and binary outcome labels. The local release contains 7,593 judgments: 5,082 training, 994 validation, and 1,517 test cases. After a shared sufficiency rule excluded 14 test records, the held-out prediction population contains 1,503 cases.

The evidence corpus is an eCourts-derived collection of Indian Supreme Court judgments covering 1950–2020, providing judgment PDFs and the metadata absent from ILDC: citations, exact decision dates, court and case identifiers, source paths, and page and character locators. The download contained 39,069 English PDF instances. After quality repair and exclusion, 39,066 PDFs yielded 2,343,435 labelled chunks, of which 2,036,981 unique chunks were loaded into a PostgreSQL provenance store and a SQLite FTS5 BM25 index.

This split follows the information actually available in each source. ILDC supports fixed-split outcome prediction but exposes only a year in the case identifier and lacks the citation and passage provenance that verification requires. The eCourts collection supports dated, traceable retrieval

TABLE II
Operational definitions used throughout the frozen evaluation.

Term	Implemented definition
Facts-only input	Text retained before the earlier of a recognised dispositive cue and a 60% character cap, sentence-aligned where possible; inputs below 10% retention or 100 words are excluded
Temporal existence	The source has a parseable exact decision date in corpus metadata
Temporal applicability	For query year Y , an authority is eligible only if its decision year is $< Y$; same-year, later-year, and missing-date sources are excluded
Temporal effectiveness	The eligibility predicate is applied inside the BM25 candidate relation before ranking and LIMIT 100, so ineligible sources cannot consume returned depth
Recall@100	Share of answer-key cases whose predefined authority occurs anywhere in the returned top-100 candidates
Recall@5	Share of answer-key cases whose predefined authority is among the five selected and displayed sources
Provenance validity	A displayed item reproduces the stored source ID, citation, decision date, court, PDF/page/character locator, exact passage, and retrieval-run membership
Citation groundedness	Every material displayed proposition is verbatim text of a supplied evidence passage and links to that evidence item
Authority consistency	A verified displayed source matches the answer-key authority by stable source ID, normalised citation, or normalised title plus exact decision date

but is not used as a substitute outcome-label benchmark. Cross-corpus links are drawn only after the alignment checks described below.

B. Source-quality audit and targeted OCR repair

A full audit found the stored text valid UTF-8, but 15 PDF instances (representing 14 source IDs) had missing or badly corrupted embedded text. Failures concentrated in image-backed Supreme Court Reports from the 1980s and 1990s, with one mojibake-affected 2018 file. Fixed checks covered absent embedded text, control characters, mojibake markers, visible-ASCII ratio, and English-token ratio.

Only flagged instances were re-rendered and processed with English Tesseract OCR at 250 DPI [8]. Twelve passed the same post-OCR quality gate and were restored; three single-page PDFs remained below threshold and were transparently excluded rather than forced into the index. Raw PDFs were preserved unchanged, exclusions were recorded, and the provenance store and BM25 index were rebuilt from the accepted corpus.

C. Cross-corpus identifier alignment

Linking ILDC query cases to eCourts sources requires establishing that two records describe the same judgment. The obvious route — converting an eCourts identifier of the form YYYY INSC N to the ILDC-like string YYYY_N — is unsafe, and quantifiably so.

That conversion produced 5,391 syntactic candidates. Only 11 passed content alignment; 5,380 were identifier-namespacing collisions. The failure is not a single offset or an isolated scrape defect: it concentrates in legacy material where similar numeric suffixes routinely denote entirely different judgments. A pipeline that trusts identifier equality here would silently merge unrelated cases, corrupting both deduplication and leakage control.

The corrected pipeline treats identifier equality as candidate generation only. A reusable gate evaluates title or party identity together with direct six-token phrase overlap before any mapping may drive deduplication or retrieval exclusion. Combining syntactic and title/party candidates yielded 8,927 deduplicated pairs, of which 1,304 were accepted and 7,623 rejected. Retrieval additionally applies a full-document self-match safeguard, so that an unmapped copy of the query judgment can be removed without suppressing an earlier authority merely quoted at length by the later judgment.

We regard this audit as independently useful. It demonstrates that mechanically similar ILDC and eCourts identifiers do not establish judgment identity, and it supplies a content-based alternative reusable across answer-key validation, development probes, leakage checks, and retrieval-time exclusion.

D. Source-verified authority answer key

The authority evaluation uses 30 ILDC fixed-test cases. Test membership was confirmed before external verification. For each accepted case the query judgment was inspected through a primary source or accepted eCourts mirror, one earlier authority was recorded independently of system retrieval, and that authority was reconciled to the corpus by stable source ID, normalised citation, or normalised title plus exact date for parallel reporter forms.

The identifier-collision discovery triggered a read-only audit of the then-current mappings: 20 passed, nine resolved sources failed direct content alignment, and one source was unresolved. One case was relinked to a content-aligned source and the other nine flagged records were replaced by new fixed-test cases, yielding a final audit of 30/30 direct-content passes.

The sample is not era-balanced: five query cases are from the 1970s, 13 from the 1980s, nine from the 1990s, two from the 2000s, and one from the 2010s. The 1980s account for 43.3%. This concentration is a consequence of the source and alignment gates rather than deliberate temporal sampling, and we treat it as a limitation (Section X).

A later additive extension verified seven further fixed-test cases under the same gates, without modifying the

original 30-case key: 1990_234, 1990_256, 1990_324, 1991_136, 1991_87, 1992_286, 1993_90. Three further candidates were held back as needing review (a background listing, an unresolved court treatment, and a party-only reliance) and are not part of any evaluation. The combined 37-case population is therefore a later expanded analysis, not the preregistered frozen base; all results below retain Base-30, Extension-7, and Combined-37 strata so the original 30-case evaluation remains independently reproducible.

V. System Design

The system comprises four auditable layers.

1. Corpus and alignment layer. ILDC split files supply prediction cases and labels. Cleaned eCourts chunks retain stable source, date, citation, court, PDF, page, and character provenance. A content-aligned crosswalk plus a runtime content check control target-case leakage.

2. Outcome branch. A shared facts-only extractor feeds a sparse linear baseline (E1) and an InLegalBERT chunk-and-pool model (E2). These baselines produce binary predictions and never retrieve external evidence.

3. Evidence branch (E3). Full facts-only text is converted to salient terms, matched through BM25 under pre-ranking temporal eligibility, and narrowed to five source-diverse passages. The renderer emits these passages in a fixed evidence-linked structure.

4. Verification and reporting layer (E4). Each displayed evidence record is verified against the corpus and the recorded retrieval run; duplicate and temporal rules are enforced; verified sources are then separately compared with the source-first answer key. Versioned JSON and Markdown artifacts preserve metrics, failures, and configuration provenance.

The answer layer is deliberately extractive. It favours inspectability over fluent synthesis: material legal text is displayed verbatim, and the conclusion is fixed, non-inferential wording tied to evidence identifiers. The renderer cannot retrieve, rerank, invent an authority, paraphrase a material proposition, or infer an outcome.

VI. Method

A. Shared facts-only input

E1 and E2 share the extraction configuration `ildc-predecision-facts-v1`. The extractor retains text preceding the earlier of a recognised dispositive cue and 60% of the document, moving the boundary to a preceding sentence end when sufficient text remains. Cases below 10% retention or 100 words are excluded. Neither prediction path uses retrieval. Model and threshold selection uses validation data only; the held-out test split is evaluated once, after selection.

B. E1: sparse linear baseline

E1 uses lowercased, Unicode-normalised TF-IDF unigrams and bigrams with sublinear term frequency, minimum document frequency two, L_2 normalisation, and

at most 100,000 features. Logistic-regression values $C \in \{0.1, 1.0, 10.0\}$ are compared by validation accuracy, with smaller C breaking ties. The selected $C = 10.0$ pipeline is refitted once on eligible training and validation cases and evaluated once on test data. Seed 202605.

C. E2: InLegalBERT chunk-and-pool

E2 fine-tunes law-ai/InLegalBERT (revision b5ecfed8) with a new two-label head. It supersedes an earlier 256-token prefix run that truncated 99.20% of eligible test inputs — a discarded result we retain for traceability. Each facts-only document is instead represented by overlapping 512-token windows with 50-token overlap. Training uses three epochs, seed 202607, learning rate 2×10^{-5} , weight decay 0.01, warm-up ratio 0.1, gradient checkpointing, and gradient accumulation.

Mean pooling of window logits before softmax is the primary rule, and checkpoint selection uses validation document-level mean-logit accuracy only. Majority vote is reported as a secondary comparison, not a post-hoc selection rule. All 1,503 eligible test documents and 9,576 test windows are covered.

D. E3: temporally constrained retrieval

The query builder `tfidf-segment-salient-terms-v1` segments the full facts-only input, removes procedural-report boilerplate, scores terms deterministically, retains section and article cues, and emits at most 32 unique terms. These query a SQLite FTS5 BM25 index backed by PostgreSQL provenance.

The frozen configuration is `week11-bm25-salient-terms-preranked-temporal-v3`. The critical design choice is placement: only judgments with decision year strictly earlier than the query year enter the candidate relation before BM25 ordering and the top-100 cutoff. Applied afterwards, the same rule would allow ineligible judgments to occupy returned depth and be discarded, shrinking the effective candidate pool; applied before, the full depth is spent on eligible material.

Candidate sources are checked against the alignment-gated crosswalk and a direct self-match rule requiring at least 100 shared six-token occurrences and 80% unique candidate-source phrase coverage. A non-learned selector then chooses up to five passages in BM25 order, with at most one passage per source.

E. E4: citation and provenance verification

E4 receives the E3 answer, the retrieval-run identifier, query identity and year, and the persisted corpus records. Each displayed item must exist in the corpus, reproduce its exact passage and provenance, belong to the recorded retrieval run, avoid the query and its aligned duplicates, and pre-date the query year. Any failure rejects the citation.

Authority consistency is evaluated separately, after verification. A fully valid citation may differ from the single

expected authority without being labelled substantively irrelevant — a distinction we return to in Section VIII-D.

F. Inference-time evidence augmentation

Outcome prediction was subsequently added to E3 and E4 by reusing the frozen E2 checkpoint without fine-tuning. A shared input builder concatenates the frozen facts extract and the selected verbatim evidence in selection order, applies the same 512-token windows with 50-token overlap, pools window logits by their mean, and selects the class by argmax. Labels, citation identifiers, retrieval scores, decision dates, provenance fields, and rendered boilerplate are excluded from model input. Seed 202607; configuration e3e4-evidence-augmented-legalbert-checkpoint6318-v1.

We note the methodological caveat plainly: the checkpoint was trained on facts-only inputs, so applying it to facts plus retrieved passages is an inference-time distribution shift, not evidence-aware fine-tuning. These predictions are therefore reported descriptively — on the frozen 30-case base and, in the RQ1 extension, on the combined 37 cases — and are not directly comparable with the 1,503-case baselines, which form a separate population with a different denominator.

G. Bundled-intervention caveat

E4 is not a single-variable ablation of E3. It jointly checks corpus existence, exact passage and provenance identity, retrieval-run membership, duplicate status, and temporal eligibility, and both stages share the same final retrieval and selected evidence. Aggregate reliability cannot be attributed to any one verification component; the supported claim concerns the complete bundle.

VII. Citation and Provenance Protocol

Each chunk in the corpus is keyed by a stable identifier built from where it originated (source path) and where it sits within that source (passage location). Against that key, the persisted record stores: source ID, source case ID where available, citation, exact decision date, court, local PDF file, page number, character start and end, and chunk text. Retrieval runs record query ID, year, query text, index version, temporal policy, rank, score, and temporal status.

The renderer may expose only supplied selected evidence. Material content remains verbatim; authority cards and conclusions refer to evidence IDs. Verification then checks corpus existence, exact field equality, run membership, duplicate status, and temporal eligibility. Answer-key matching occurs only after a displayed citation has passed these checks. Parallel reporter forms are reconciled through stable source identity or normalised title plus exact decision date, never loose string similarity.

This protocol distinguishes four failure states that a single “correct citation” label would collapse:

- 1) a fabricated or altered citation;

TABLE III
Outcome prediction. Populations are distinct and must not be compared as a like-for-like leaderboard.

Model	n	Acc.	Macro-F1
Majority class	1,503	0.5017	0.3341
E1 TF-IDF + LR	1,503	0.6134	0.6123
E2 InLegalBERT, mean logits	1,503	0.5968	0.5924
E2 InLegalBERT, maj. vote	1,503	0.6015	0.5937
E3 evidence-augmented	30	0.6667	0.6032
E4 verified evid.-augmented	30	0.6667	0.6032

- 2) a real citation that was not retrieved for this query;
- 3) a retrieved citation violating temporal or duplicate policy;
- 4) a fully valid citation that does not match the predefined expected authority.

Keeping these separate is what allows failures to be located rather than merely counted. In our results, states (1)–(3) occur zero times and state (4) occurs in 18 of 30 cases — a profile that a single aggregate score would render invisible.

VIII. Results

A. Reporting populations

Four populations are reported and never aggregated across kinds: 1,503 eligible fixed ILDC test cases for outcome prediction; a combined 37-case reference-evidence population for authority recovery, evidence-augmented prediction, and 185 displayed-citation verifications, always stratified as Base-30 (the frozen key: 150 citations), Extension-7 (the additive verified cases: 35 citations), and Combined-37; and paired cases for explanation-format review (seven formative author-reviewed pairs, superseded as the principal RQ3 result by the 14-case LLM exploratory evaluation below).

B. Outcome prediction

Table III reports the outcome results. On the 1,503-case population, the sparse E1 baseline reached accuracy 0.6134 and macro-F1 0.6123, exceeding corrected E2 mean-logit pooling at 0.5968 and 0.5924. E2’s secondary majority-vote aggregation reached 0.6015 and 0.5937. All primary models exceeded the 0.5017 majority-class accuracy.

This result does not show that domain pre-training is ineffective in general. It shows that under the implemented facts extraction, data size, training budget, window aggregation, and frozen settings, the sparse baseline performed better. The per-case analysis in Section IX shows the two models make substantially non-identical errors.

On the separate 30-case subset, evidence-augmented E3 and verified E4 both reached accuracy 0.6667 and macro-F1 0.6032, with identical confusion matrices. On the combined 37 cases the same shared predictor reaches accuracy 0.6486 and macro-F1 0.6073 with confusion

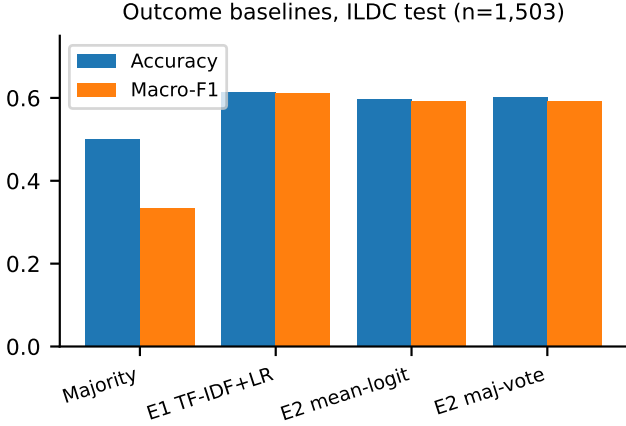


Fig. 1. Accuracy and macro-F1 for E1, corrected E2 under both pooling rules, and the majority baseline on the eligible ILDC test population ($n = 1,503$).

matrix $[[6,3],[10,18]]$, again identical for E3 and E4 (Table IV). The equality is expected and is not evidence that verification is redundant. E4 verification did not reject any evidence in this run, so the predictor received an identical input; E4’s contribution is control, traceability, and explanation, not a different decision rule. A configuration in which verification rejected evidence would produce differing scores, and the architecture reports both so that such a divergence would be visible.

C. Authority recovery and evidence integrity

On the 30-case answer-key subset, expected-authority Recall@5 was 0.40 (12/30) and Recall@100 was 0.50 (12 selected + 3 retrieved-but-unselected = 15/30 found within $k=100$). The remaining 15 authorities were absent at $k=100$.

The principal RQ1 result uses the combined 37-case population (Table IV). The seven extension cases contribute no additional top-5 recovery (0/7) but five further top-100 recoveries at ranks 14, 16, 26, 32, and 89, with two absent; their displayed citations are likewise fully verified (35/35). The combined figures are therefore Recall@5 $12/37 = 0.324324$ and Recall@100 $20/37 = 0.540541$ over 185 displayed citations. The lower combined Recall@5 is descriptive of the added stratum, not evidence about the base configuration, and the frozen 30-case result above is retained unchanged alongside it.

Figure 2 shows the recovery funnel.

All 150 displayed citations were grounded, provenance-valid, and temporally eligible, with zero unsupported claims across the 30 cases (Table V, Figure 3). The final candidate log contained 3,000/3,000 eligible candidates. In a five-case explanation-faithfulness check, every evidence reference resolved to the exact persisted retrieved chunk. Over the combined 37 cases the same profile holds at

TABLE IV
RQ1 retrieval and evidence-augmented prediction by stratum. The combined row is the principal evaluation; strata are never pooled silently.

Population	n	R@5	R@100	E3 Acc.	E3 Macro-F1
Base-30	30	0.40 (12/30)	0.50 (15/30)	0.6667	0.6032
Extension-7	7	0.00 (0/7)	0.7143 (5/7)	0.5714	0.5714
Combined-37	37	0.3243 (12/37)	0.5405 (20/37)	0.6486	0.6073

TABLE V
Evidence recovery and integrity on the frozen Base-30 answer-key stratum ($n = 30$ queries; 150 displayed citations). Combined-37 values are reported in Table IV and the surrounding text.

Measure	Value	Arithmetic
Recall@5	0.40	12 of 30 in top 5
Recall@100	0.50	15 of 30 in top 100
Absent at $k=100$	0.50	15 of 30
Authority precision	0.08	12 of 150 shown
struct. ceiling	0.20	30 of 150 (5 shown, 1 credited)
Grounding	1.00	150/150 passed
Provenance validity	1.00	150/150 passed
Temporal violations	0	0 of 150
Unsupported claims	0	0 of 30 cases

larger denominators: 185/185 grounded and provenance-valid, 0/185 temporal violations, 0/37 unsupported claims.

E4 verification is fail-closed: any failing check rejects the citation and no verified output is produced for that case. On live data no citation was rejected, which is a ceiling effect of already-valid selected evidence, not a measure of the gate’s strength. Three positive controls through the unchanged verifier demonstrate that strength directly: altered verbatim passages were rejected 37/37, fabricated authority citations 37/37, and backdated query years rendering authorities ineligible 37/37.

This is the paper’s central result. Verification success and authority recovery are dissociated: the system verified everything it displayed (185/185 combined; 150/150 on the frozen base) while recovering the expected authority in 20 of 37 cases at $k=100$ (15/30 base). Neither number alone describes the system. A reviewer told only that all displayed citations are valid would over-trust it; a reviewer told only that Recall@100 is 0.54 would miss that nothing displayed was fabricated, misattributed, or anachronistic.

D. Interpreting authority-consistency precision

Authority-consistency precision was 0.08, with recall 0.40 and F1 0.133. This low absolute value must be read structurally rather than as a fabrication rate. The denominator is 150 displayed citations; the answer key credits exactly one authority per case while the renderer displays five. Perfect inclusion of the single credited authority in every case would therefore cap this measure at $30/150 = 0.20$. The observed 0.08 is 40% of that ceiling.

The 138 non-matching displayed citations are not independently annotated for substantive legal relevance and

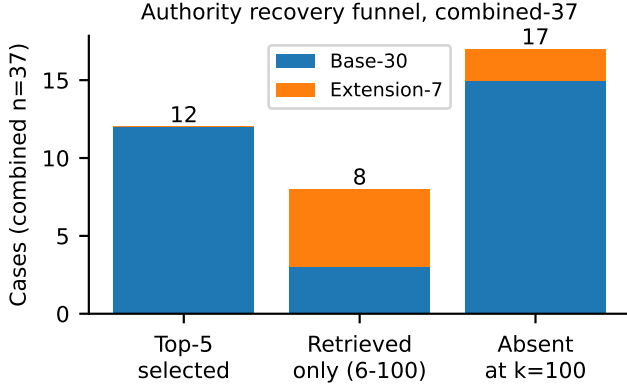


Fig. 2. Expected-authority recovery on the combined reference set ($n = 37$): 12 retrieved and selected, eight retrieved but unselected (three base, five extension), and 17 absent at $k=100$.

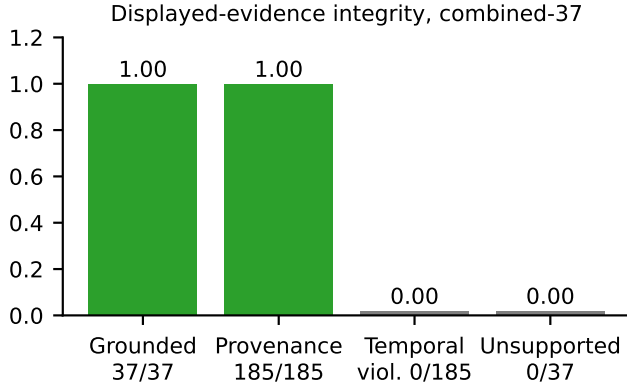


Fig. 3. Displayed-evidence integrity under the final configuration (combined $n = 37$ queries; 185 displayed citations): all grounding and provenance checks passed, with no temporal violations or unsupported claims. The frozen Base-30 stratum contributes 150/150 of these checks.

cannot be declared incorrect. They passed every integrity check; they simply are not the one authority the key records. Reporting this metric without its ceiling would materially misrepresent the system.

E. Retrieval mechanism

The final configuration was reached through three bounded changes, each validated on the nine-case development probe before being tested on the 30-case held-out set. Replacing the legacy first-32-term query with terms drawn from the full input took the probe from 0/9 to 3/9 at $k=100$. Adding an 80% source-coverage condition to the self-match guard took it to 6/9 and forced withdrawal of an earlier, overly broad lexical-mismatch claim. Moving the earlier-year requirement so that it filters candidates before BM25 ranking, rather than after, took the probe to 7/9; carried over to the 30-case held-out set, this last change alone lifted Recall@5 from 5/30 to

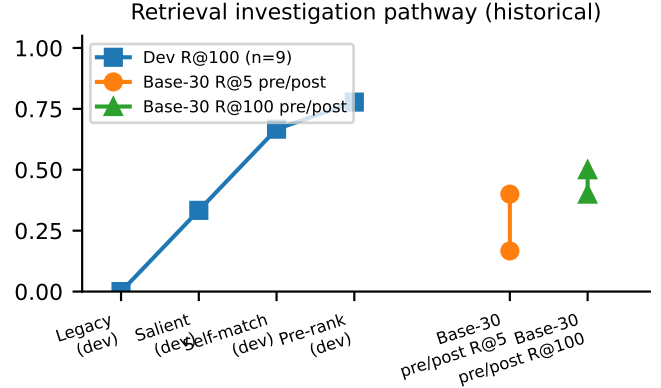


Fig. 4. Retrieval investigation pathway. The nine-case development probe covers the salient-query and self-match repairs; the separate held-out comparison ($n = 30$) covers the move from post-ranking to pre-ranking temporal filtering. These are two distinct populations, not one learning curve.

12/30 and Recall@100 from 12/30 to 15/30, and it did so without losing any of the 12 top-100 successes the prior configuration already had.

Temporal capacity is not the whole story. Case 2013_35 remained absent despite 79 eligible candidates in its earlier raw top 100, whereas 1980_105 was recovered despite only three eligible candidates and became a top-five hit. Eligible-pool size and lexical relevance are independent contributors, and residual lexical mismatch remains the dominant unresolved cause.

F. Explanation format

The principal RQ3 evaluation is a blinded exploratory comparison by four LLM raters (Gemini, Claude Sonnet 4.6, DeepSeek, GPT-5.6 Luna) over 14 paired cases with citations held programmatically constant in shuffled order: the seven formative pairs plus all seven verified extension cases, with no cherry-picking. Each rater scored five transparency dimensions on a 1–5 scale and recorded a forced preference, yielding 112 display-level ratings and 56 preferences. The structured presentation was preferred in 56/56 evaluations, with mean differences of +1.75 (evidence linkage), +1.00 (citation verifiability), +2.48 (traceability), +2.52 (explanation clarity), and +1.98 (transparency/inspectability), for an overall +1.95 (4.66 vs. 2.71); see Figure 5.

This is explicitly not human evaluation: no independent rater participated, no significance test was run, and the result establishes nothing about human preference or legal correctness. Two further cautions apply. First, the DeepSeek and GPT-5.6 Luna runs produced byte-identical rating vectors; both are retained and reported separately rather than merged, so the four-rater unanimity partly reflects duplicated outputs. Second, the earlier seven-case author self-review (means 4.57 vs. 2.71, 4.57 vs. 2.29, 4.43 vs. 2.86, 4.43 vs. 3.00; 7/7 structured) is kept only as

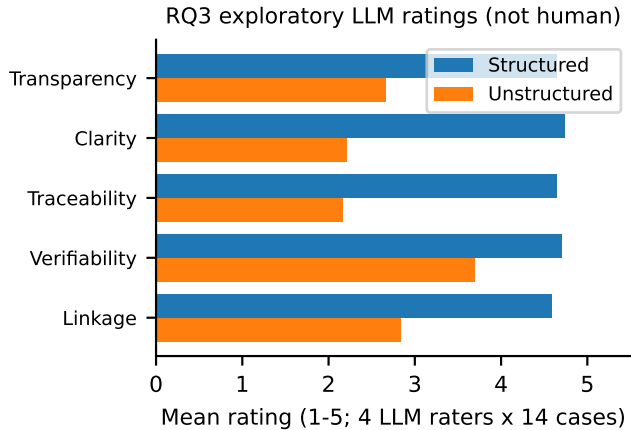


Fig. 5. Blinded LLM-rater means by dimension ($n = 14$ cases $\times$ 4 raters). This is exploratory presentation evaluation, not independent human-subject evidence.

TABLE VI
Outcome-error overlap by evaluation population. “E1” and “E2” count cases where only that model was correct.

Population	Both	E1	E2	Neither
Full test ($n = 1,503$)	684	238	213	368
Answer key ($n = 30$)	18	3	3	6

formative, non-independent evidence, superseded by the 14-case evaluation above. Its most useful output remains negative: generic uncertainty wording did not adapt to the evidence quality of 2013_35, indicating uncalibrated caution language that a purely quantitative evaluation would not have surfaced.

IX. Error Analysis

A. Outcome disagreement

The analysis is read-only and uses only final frozen outputs; reconstructed E1 and inference-only E2 predictions exactly reproduced their reported metrics before per-case joining. Across 1,503 cases, both models were correct on 684 and both wrong on 368; E1 alone was correct on 238 and E2 alone on 213 (Table VI). The models therefore make complementary errors, even though E1 leads in aggregate — a pattern that suggests ensemble or disagreement-triggered routing as future work rather than a straightforward verdict against domain pre-training.

In the evidence-augmented run, E2 was wrong while E3 and E4 were correct for two cases (1974_36, 1984_136). E3-correct/E4-wrong and E3-wrong/E4-correct each occurred in 0/30 cases, because verification changed neither the selected evidence nor the shared predictor input.

Critically, four cases (2008_1629, 1981_187, 1982_29, 1985_40) retrieved and selected the expected authority yet still produced an incorrect outcome prediction. Correct authority recovery does not mechanically imply a correct

TABLE VII
Exhaustive authority-recovery partition ($n = 30$).

Outcome	n	Cases
Retrieved and selected	12	2008_1629, 1995_322, 1995_375, 1986_176, 1977_99, 1981_187, 1980_222, 1980_105, 1995_425, 2002_944, 1982_29, 1988_96
Retrieved, not selected	3	1980_133 (rank 15), 1981_55 (rank 28), 1985_40 (rank 78)
Absent at $k=100$	15	1997_792, 1993_185, 1971_295, 1974_36, 1986_378, 1984_136, 2013_35, 1980_217, 1978_33, 1977_145, 1995_412, 1995_403, 1986_397, 1994_632, 1992_84

predicted outcome — further support for keeping the two measures apart. No correct prediction was accompanied by an unsupported citation.

B. Authority-recovery buckets

Table VII gives the exhaustive partition. The middle bucket isolates selection failure: the authority was available in the candidate set but fell outside the five displayed sources, which is addressable by a better selector under the existing retrieval. The absent bucket requires improvement upstream, in query representation, lexical matching, authority-aware ranking, or corpus coverage.

C. Worked example: retrieval noise made visible

Case 1985_40 illustrates why extractive presentation aids review. The facts concern a factory at Atul manufacturing dyes, chemicals, and pharmaceuticals, and the addition of synthetic organic dyestuffs as item 14D to the First Schedule of the Central Excise and Salt Act, 1944.

The system displayed five verified citations: [1984] 3 S.C.R. 930, [1972] 3 S.C.R. 770, [1980] 3 S.C.R. 1109, [1971] 3 S.C.R. 506, and [1973] 1 S.C.R. 822. The expected authority, [1964] 2 S.C.R. 888, was retrieved at rank 78 but not selected.

Inspecting the displayed set, the first concerns excise valuation and the “related person” definition for Atul Products, but the second concerns an industrial dispute at Atul, the third rayon and silk sales tax, and the fifth cigarette excise duty. The separate authority index and per-evidence blocks made these apparent subject-matter mismatches visible for review in a way continuous prose did not. We stress that this is an illustration of navigability, not an independent annotation that the alternative citations are legally irrelevant.

X. Limitations and Threats to Validity

Scope. The study covers English-language Indian Supreme Court judgments only. It does not evaluate other Indian languages, High Court or lower-court material, separately versioned statutes, or cross-jurisdictional transfer.

Extraction approximation. Targeted OCR restored 12 of 15 affected PDF instances; three were excluded and repaired text may retain minor noise. ILDC provides no gold facts/reasoning boundary, so the deterministic facts rule can retain some reasoning or omit facts near a boundary.

Answer-key size and era concentration. The evidence evaluation rests on 30 source-verified frozen cases, extended additively to 37 with seven later verified cases that postdate the original freeze; 13 of the base 30 are from the 1980s. The combined set is neither large nor era-balanced and should not be read as representative of the ILDC test set or of Indian legal research generally. This is the primary constraint on the generality of the retrieval findings.

Single reference authority. The key records one expected authority per query, not every relevant source. Authority inconsistency is a reproducible reference mismatch, not a substantive legal-relevance judgment.

Residual recovery gap. At $k=100$, half of the expected authorities still went unrecovered. Comparing 2013_35 against 1980_105 rules out an easy explanation: having enough temporal room for the true authority to qualify is not, by itself, sufficient for the retriever to surface it, so the remaining gap has to be attributed to ranking quality rather than to the eligibility window.

Year-level temporal granularity. ILDC supplies query dates at year-level precision only. As a result, any candidate sharing a query’s year is excluded rather than risk admitting a source that actually postdates it – a conservative choice that can discard genuinely eligible evidence. Getting day-level dates would require manually checking each case against a primary source, since ILDC’s public release does not provide them; the limitation therefore sits in the dataset, not in how the pipeline parses dates.

Bundled verification. E4 evaluates several integrity checks together and cannot isolate the causal contribution of any single component.

Non-independent explanation review. The seven-case review is a non-random author self-review, not independent usability or legal-correctness evidence. The 14-case LLM comparison that supersedes it is exploratory presentation evaluation by four model raters, not human-subject evidence; two of those runs produced identical outputs and are reported separately rather than merged, so even the unanimous preference partly reflects duplicated runs.

Prototype scale. Results apply to the frozen implementation and samples rather than to all users, courts, domains, or changing corpora.

XI. Responsible Use and Governance

The system is designed as a legal-research aid, not an adjudicator or a provider of legal advice. Although E3 and E4 emit experimental outcome predictions, the controlled

renderer states no legal conclusion beyond supplied evidence, and the prediction is held in a separate field. A human reviewer remains responsible for deciding whether retrieved material is relevant, current, authoritative, and applicable. This allocation follows the concern in Pooja Ramesh Singh that AI may assist legal work but cannot displace human verification and responsibility [6].

Risk controls are fail-closed. Evidence with missing dates, same-year ambiguity, later dates, target identity, or audited duplication is not displayed. A citation must resolve to exact corpus text and the correct retrieval run; altered or unsupported material is rejected. Stable source and passage locators make verification possible without trusting the renderer’s prose.

Governance also requires transparent negative results. We preserve the discarded 256-token run, superseded retrieval configurations, the identifier-collision correction, answer-key replacements, OCR exclusions, and the non-independent status of the explanation review. Evaluation populations and denominators remain separate, and alternative citations are not called irrelevant without annotation.

The source corpora contain public judgments, but public availability does not remove privacy or misuse concerns. This prototype exposes no end-user search service and automates no consequential decision. Any deployment would require access controls, updated source and licensing review, privacy assessment, independent legal evaluation, monitoring for corpus change, and a defined correction process.

XII. Conclusion and Future Work

We built and evaluated a provenance-preserving, temporally constrained evidence workflow for historical Indian Supreme Court research, and kept evidence and prediction measurements separate throughout.

The principal finding is a dissociation. Every one of 185 displayed citations across the combined 37 cases passed grounding, provenance, and temporal verification, with zero unsupported claims across the 37 cases, while 17 of the 37 independently verified expected authorities were absent at $k=100$. Verification succeeded for what the system displayed; it did not guarantee that the system found the expected authority. Legal-research systems therefore benefit from explicit temporal, identity, and provenance controls — but those controls must be evaluated alongside, never instead of, retrieval coverage and human review.

The cross-corpus audit yields a transferable methodological lesson: only 11 of 5,391 syntactic identifier matches between ILDC and the eCourts collection were content-aligned. Identifier equality is a candidate hint requiring content validation, not evidence of judgment identity.

The supporting results are reported with their limits. On the 1,503-case outcome task the sparse TF-IDF baseline outperformed corrected InLegalBERT under the

frozen settings, a result about this experiment rather than a general rejection of legal-domain pre-training. Inference-time evidence augmentation reached 0.6667 accuracy and 0.6032 macro-F1 on the frozen 30-case subset (0.6486 and 0.6073 on the combined 37), descriptively and under acknowledged distribution shift. The 14-case structured-presentation comparison is exploratory LLM-rater evidence, not human preference; the seven-case self-review beneath it is formative and non-independent.

Future work. The highest-value extension is an era-balanced answer key of 75–100 cases, large enough to support a paired significance test such as McNemar’s test between retrieval configurations. A blinded review with independent raters and reported inter-annotator agreement (e.g., Fleiss’ κ) would replace the formative explanation evidence. A hybrid dense–sparse retrieval comparison would test whether semantic retrieval closes the residual lexical misses while retaining the present temporal and provenance controls, and evidence-sensitive uncertainty language would address the calibration weakness the review exposed.

Reproducibility

All configurations, seeds, model revisions, corpus identities, and evaluation artifacts are frozen and machine-readable. The E2 checkpoint (checkpoint-6318) is identified by SHA-256 924a5bb9...dbdc773; the evidence-augmented replay produces a stable output hash. The implementation passes 81 automated tests and the documented reproducibility checks (14/14 documentation checks and 34/34 freeze-hash checks in the Week-16 audit), and the evidence-augmented evaluation replays deterministically from a clean clone. Code and artifacts are available at <https://github.com/0-YuvrajSingh/NyayaTrace>.

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